

Size statistics of cumulus cloud populations in large-eddy simulations

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How many clouds of each size are there, and which sizes carry the cloud cover and the vertical mass flux?



Aims

- 1 Test LES cloud populations against observed ones, with the algorithm of satellite studies.
- 2 Find a universal form of the cloud size density.
- 3 Find which sizes dominate cover and mass flux.

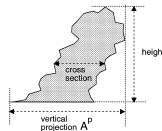
Simulations: KNMI LES

$\Delta x = \Delta y = 50$ m, $\Delta z = 40$ m, 6.4 km domain;
3D fields sampled every 5 min.

BOMEX	ocean, trade-wind cumulus, steady	36,776 clouds
SCMS	Florida, land, deepening layer	20,268 clouds
ARM SGP	Oklahoma, land, diurnal cycle	35,137 clouds

Cloud statistics

Cloud n : connected saturated grid boxes, sized by the projected area A_n^p (right).



$$\ell_n = \sqrt{A_n^p}, \quad N = \int_0^\infty \mathcal{N}(l) dl$$

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y}, \quad a^p = \int_0^\infty \alpha^p(l) dl$$

$$\mu(l) = \frac{1}{h_c} \int_{h_c} \alpha(l, z) w(l, z) dz, \quad m = \int_0^\infty \mu(l) dl$$

$$\mathcal{N}(l) = a l^b, \quad \mathcal{N}^*(\log l) = l \ln 10 \mathcal{N}(l) \propto l^{b+1}$$

$\mathcal{N}^*$: logarithmic bins, log-log slope $b + 1$. $\alpha(l, z)$: cloud fraction at height z ; w : cloud-mean vertical velocity; h_c : cloud-layer depth.

LES clouds are measured exactly as satellites measure real ones

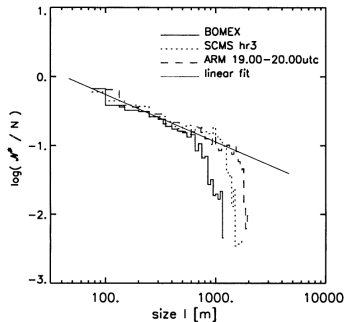


Fig. 4a: N^*/N for the three cases; fit $1.121 - 0.70 \log l$ below the break.

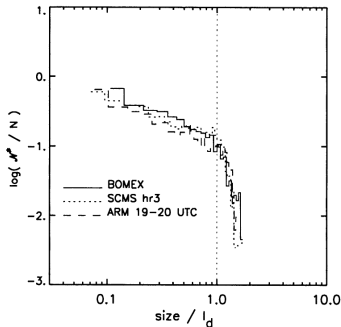


Fig. 6: the same densities against l/l_d ; dotted line at the break.

Below the break

$b = -1.70$ (log-bin slope -0.70) over about one decade, from the grid spacing to ℓ_d .

Observed: log-bin slope -0.89 (Cahalan & Joseph 1989); mean $b = -1.98$ (Benner & Curry 1998).

Scale break ℓ_d

BOMEX 700 m

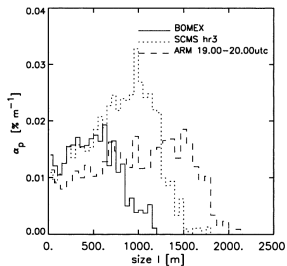
SCMS 1,050 m

ARM, 15-16 UTC 400 m

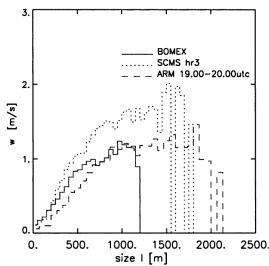
ARM, 19-20 UTC 1,250 m

More sampled clouds: the same break.

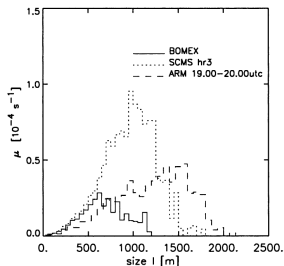
One exponent fits every case; only the break ℓ_d changes



(a) projected cover $\alpha^p(l)$



(b) cloud-mean $w(l)$



(c) mass flux $\mu(l)$

Pure power law: $\alpha^p(l) \propto l^{b+2}$; the smallest clouds dominate if $b < -2$, the largest if $b > -2$. Here $b = -1.70$ below ℓ_d , $b < -2$ above: α^p peaks near ℓ_d .

$\mu = \alpha w$ peaks at larger sizes, since w grows with l . The smallest clouds carry almost no mass flux.

Robustness

Grid 25–100 m:
same dominant and largest sizes.

Domain 3.2 km: the largest clouds are missing; dominance moves to the largest size.

Shear $\times 0, 1, 2$:
broader density, larger ℓ_d ; α^p from 13 to 19%, while $m \approx 0.015 \text{ m s}^{-1}$ in all three.

The dominant size sits at the break, far above the grid scale, for cover and for transport

Limits

Stated in the paper

One decade of scaling; 1–2 m grids needed to extend it.

The origin of the break is unexplained.

Small domains truncate the largest clouds.

Three cases; exponents compared with published values, not the same scenes.

Our reading [ours]

Snapshots: no cloud life cycle.

No error bar on b ; least squares on binned histograms.

Only saturated air counts: dry thermals are invisible.

In the paper

Large clouds root in the subcloud layer.

Suggested regimes: above ℓ_d , coherent structures of subcloud turbulence; below, large clouds decaying into smaller ones.

Other ideas: the largest boundary-layer cells (Cahalan & Joseph 1989); subcloud humidity (Jonker et al. 1999).

ARM: ℓ_d grows from 400 to 1,250 m as the (sub)cloud layer deepens.

Proposal, not in the paper

1. Same pipeline on climbs: patches of climb crossings on a height plane, $\ell = \sqrt{\text{area}}$, $\mathcal{N}(\ell)$ in log bins. Power law? Break? Collapse by boundary-layer depth?

2. Does the patch scale grow through the day, like ℓ_d in ARM?

3. Analogue of $\mu(\ell)$: climb rate $\times$ area against size.

Caveat: pilots select strong thermals and cross them sparsely.

Climb data could test the size statistics found for clouds